Milestone3_Code_Malekshahi

April 5, 2026

1 Milestone 3 - Credit Risk Analysis Project

2 Predicting Credit Risk for Loan Approval Decisions

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[1]:
```
# import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import (
    accuracy_score,
    classification_report,
    confusion_matrix,
    ConfusionMatrixDisplay
)
```

[2]:
```
# Load the dataset

file_path = "german_credit_data.csv"

df = pd.read_csv(file_path)

print("First 5 rows:")
display(df.head())

print("\nShape of dataset:")
print(df.shape)
```

```python
print("\nColumn names:")
print(df.columns.tolist())

print("\nData types:")
print(df.dtypes)

print("\nMissing values:")
print(df.isnull().sum())
```

```
First 5 rows:

   Unnamed: 0  Age     Sex  Job Housing Saving accounts Checking account  \
0           0   67    male    2     own             NaN           little
1           1   22  female    2     own          little         moderate
2           2   49    male    1     own          little              NaN
3           3   45    male    2    free          little           little
4           4   53    male    2    free          little           little

   Credit amount  Duration              Purpose  Risk
0           1169         6             radio/TV  good
1           5951        48             radio/TV   bad
2           2096        12            education  good
3           7882        42  furniture/equipment  good
4           4870        24                  car   bad


Shape of dataset:
(1000, 11)

Column names:
['Unnamed: 0', 'Age', 'Sex', 'Job', 'Housing', 'Saving accounts', 'Checking
account', 'Credit amount', 'Duration', 'Purpose', 'Risk']

Data types:
Unnamed: 0           int64
Age                  int64
Sex                 object
Job                  int64
Housing             object
Saving accounts     object
Checking account    object
Credit amount        int64
Duration             int64
Purpose             object
Risk                object
dtype: object

Missing values:
```

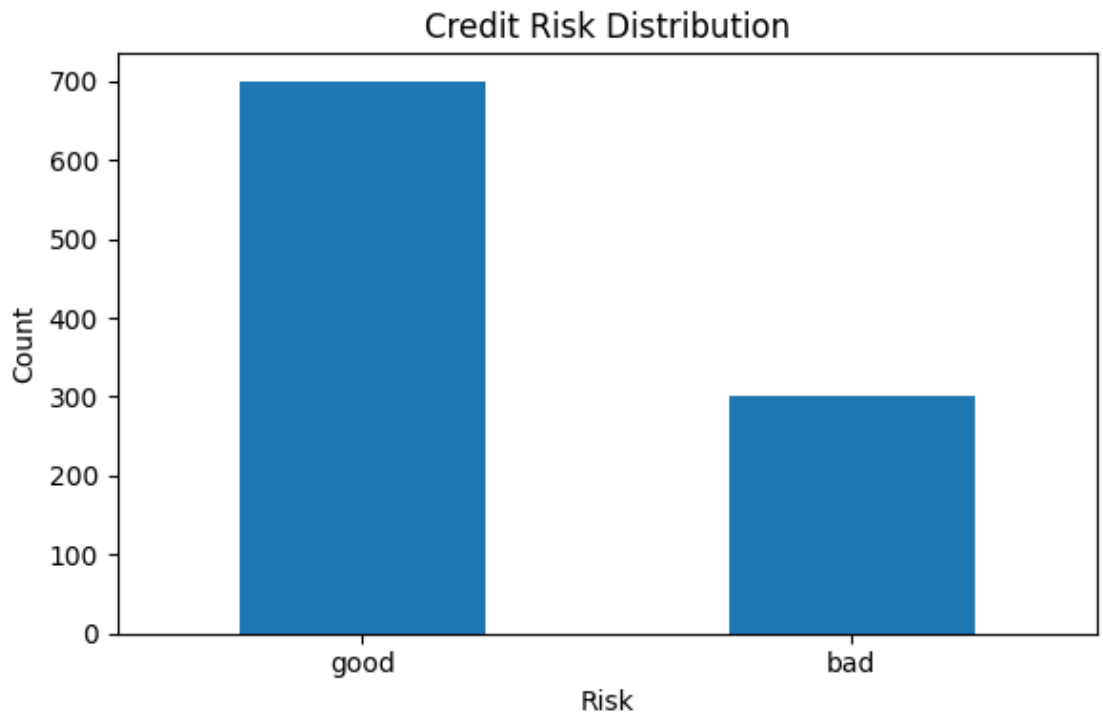
```
Unnamed: 0            0
Age                   0
Sex                   0
Job                   0
Housing               0
Saving accounts     183
Checking account    394
Credit amount         0
Duration              0
Purpose               0
Risk                  0
dtype: int64
```

[3]:
```python
# Basic cleaning
if "Unnamed: 0" in df.columns:
    df = df.drop(columns=["Unnamed: 0"])

print("\nUpdated shape after cleanup:")
print(df.shape)

print("\nUnique values in Risk:")
print(df["Risk"].value_counts())
df.columns = df.columns.str.strip()
```

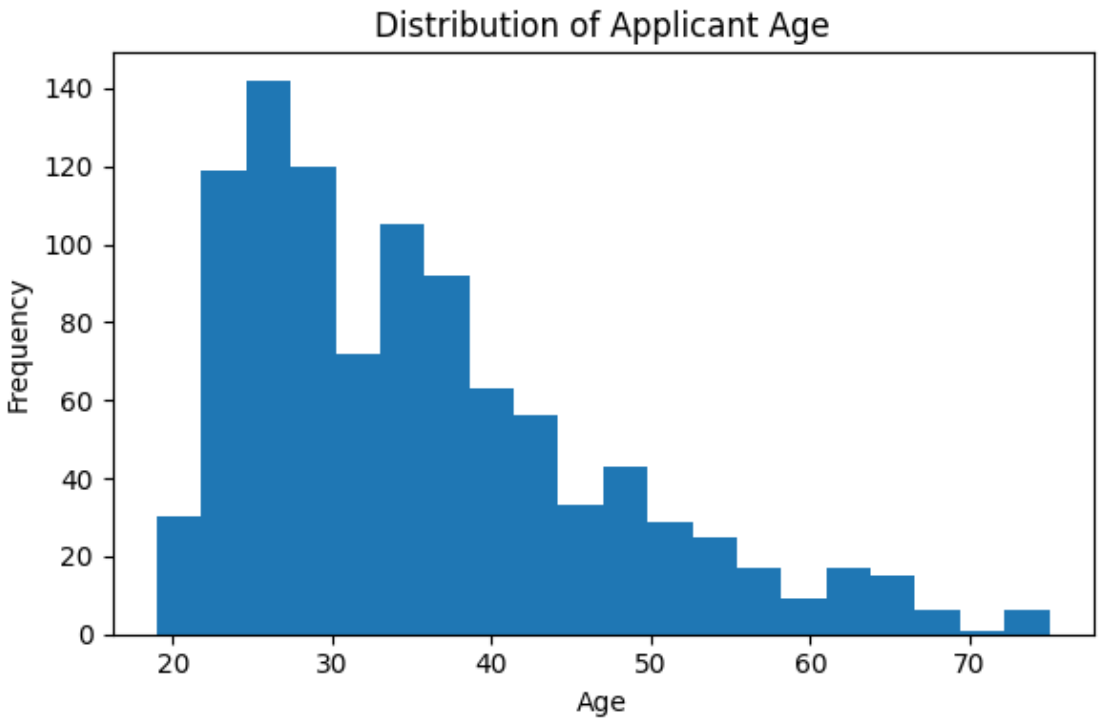
```
Updated shape after cleanup:
(1000, 10)

Unique values in Risk:
Risk
good    700
bad     300
Name: count, dtype: int64
```

2.1 Exploratory Data Analysis (EDA)

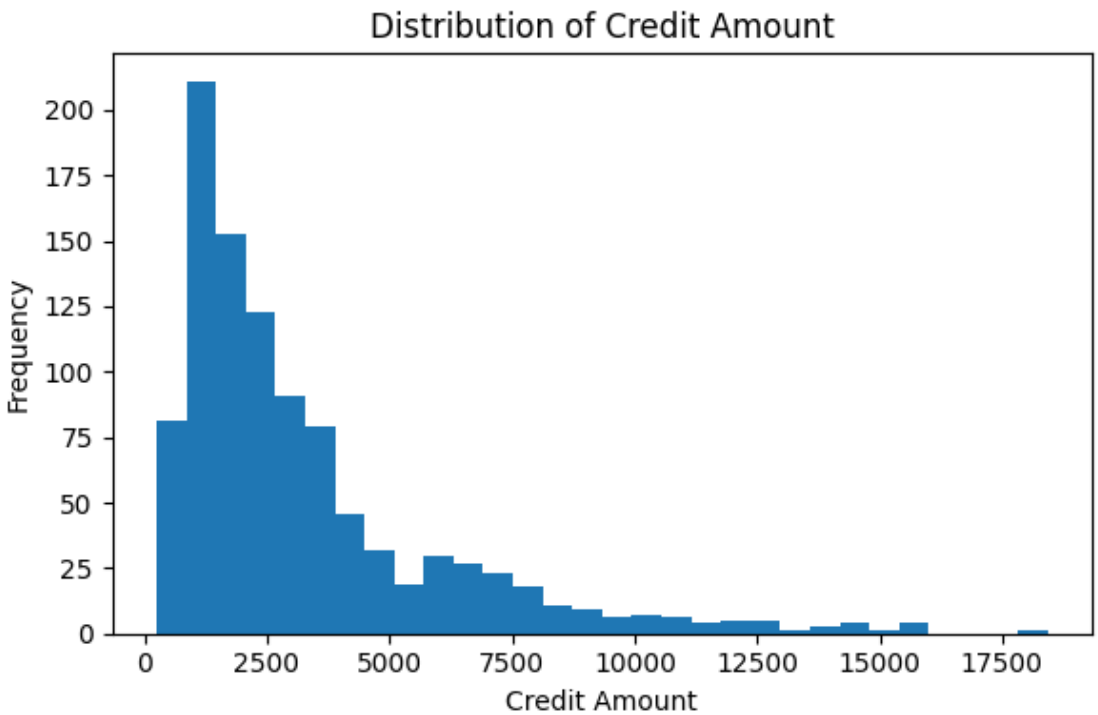
[4]:
```python
# a. Target distribution
plt.figure(figsize=(6, 4))
df["Risk"].value_counts().plot(kind="bar")
plt.title("Credit Risk Distribution")
plt.xlabel("Risk")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



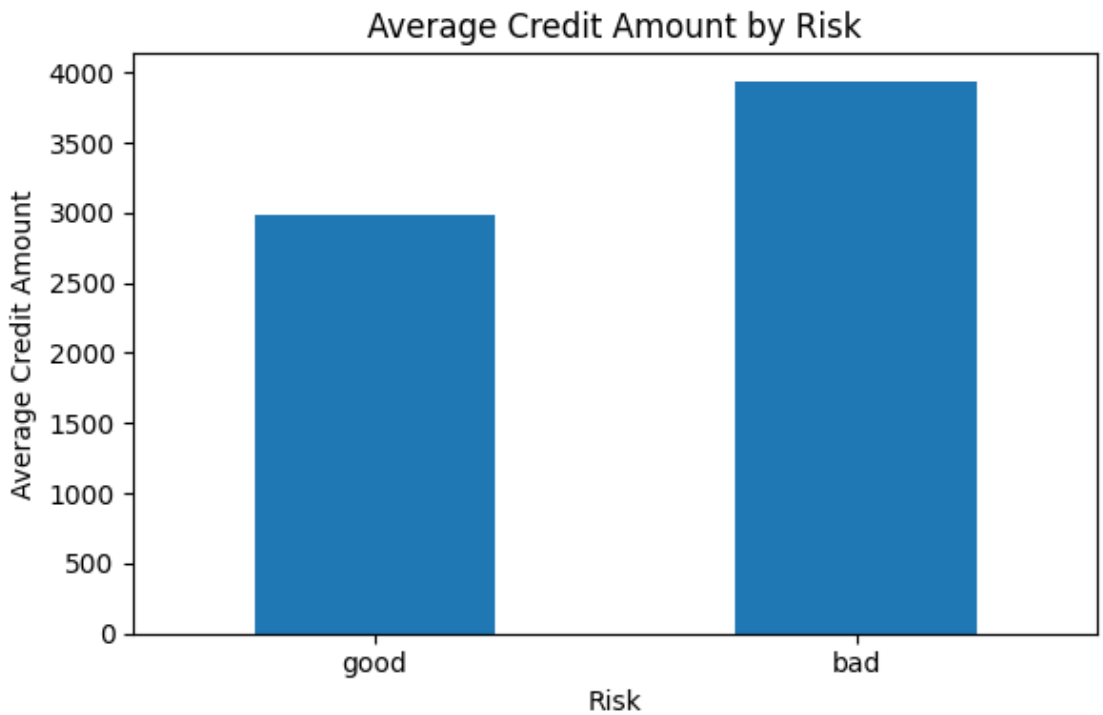
[5]:
```
# b. Distribution of age
plt.figure(figsize=(6, 4))
plt.hist(df["Age"], bins=20)
plt.title("Distribution of Applicant Age")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



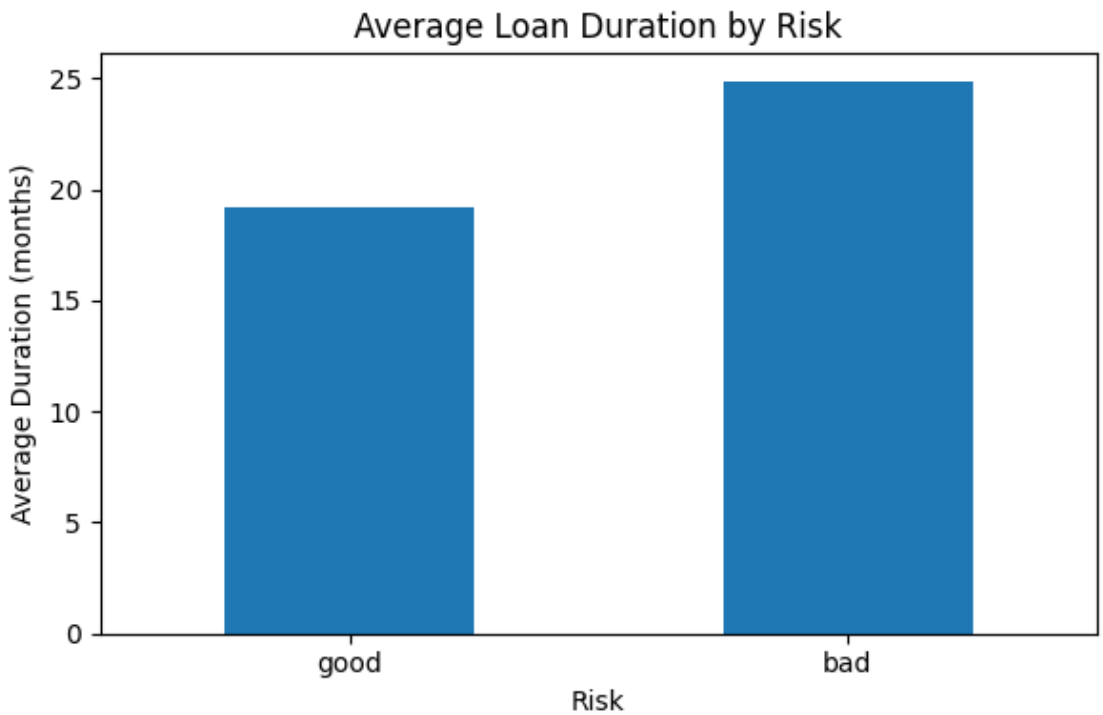
[6]:
```
# c. Distribution of credit amount
plt.figure(figsize=(6, 4))
plt.hist(df["Credit amount"], bins=30)
plt.title("Distribution of Credit Amount")
plt.xlabel("Credit Amount")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



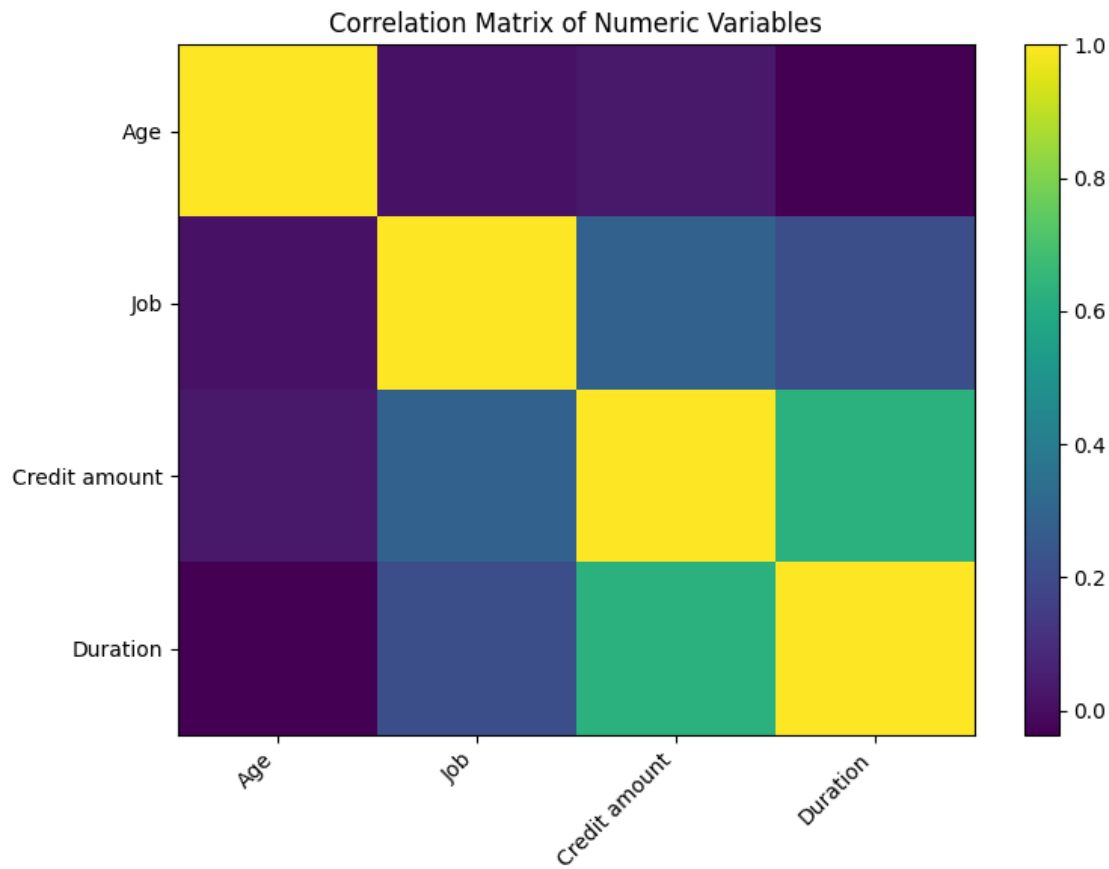
[7]:
```
# d. Average credit amount by risk
avg_credit_by_risk = df.groupby("Risk")["Credit amount"].mean().sort_values()

plt.figure(figsize=(6, 4))
avg_credit_by_risk.plot(kind="bar")
plt.title("Average Credit Amount by Risk")
plt.xlabel("Risk")
plt.ylabel("Average Credit Amount")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



[8]:
```
# e. Average loan duration by risk
avg_duration_by_risk = df.groupby("Risk")["Duration"].mean().sort_values()

plt.figure(figsize=(6, 4))
avg_duration_by_risk.plot(kind="bar")
plt.title("Average Loan Duration by Risk")
plt.xlabel("Risk")
plt.ylabel("Average Duration (months)")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



[9]:
```
# f. Numeric correlation matrix
numeric_cols = df.select_dtypes(include=["int64", "float64"]).columns.tolist()

corr = df[numeric_cols].corr()

plt.figure(figsize=(8, 6))
plt.imshow(corr, interpolation="nearest", aspect="auto")
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=45, ha="right")
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title("Correlation Matrix of Numeric Variables")
plt.tight_layout()
plt.show()

print("\nSummary statistics:")
display(df.describe(include="all"))
```




```
Summary statistics:

                Age   Sex          Job Housing Saving accounts  \
count   1000.000000  1000  1000.000000    1000              817
unique          NaN     2          NaN       3                4
top             NaN  male          NaN     own           little
freq            NaN   690          NaN     713              603
mean      35.546000   NaN     1.904000     NaN              NaN
std       11.375469   NaN     0.653614     NaN              NaN
min       19.000000   NaN     0.000000     NaN              NaN
25%       27.000000   NaN     2.000000     NaN              NaN
50%       33.000000   NaN     2.000000     NaN              NaN
75%       42.000000   NaN     2.000000     NaN              NaN
max       75.000000   NaN     3.000000     NaN              NaN

       Checking account  Credit amount     Duration Purpose  Risk
count               606    1000.000000  1000.000000    1000  1000
unique                3            NaN          NaN       8     2
top              little            NaN          NaN     car  good
```

```
freq                274            NaN          NaN     337   700
mean                NaN    3271.258000    20.903000     NaN   NaN
std                 NaN    2822.736876    12.058814     NaN   NaN
min                 NaN     250.000000     4.000000     NaN   NaN
25%                 NaN    1365.500000    12.000000     NaN   NaN
50%                 NaN    2319.500000    18.000000     NaN   NaN
75%                 NaN    3972.250000    24.000000     NaN   NaN
max                 NaN   18424.000000    72.000000     NaN   NaN
```

2.2 Prepare data for modeling

[10]:
```
# Convert target variable to binary
# good = 0, bad = 1
df["Risk"] = df["Risk"].map({"good": 0, "bad": 1})

print("\nMapped target values:")
print(df["Risk"].value_counts())

X = df.drop("Risk", axis=1)
y = df["Risk"]

# Identify column types
categorical_features = X.select_dtypes(include=["object"]).columns.tolist()
numerical_features = X.select_dtypes(include=["int64", "float64"]).columns.
 ↪tolist()

print("\nCategorical features:")
print(categorical_features)

print("\nNumerical features:")
print(numerical_features)

# Preprocessing for numeric columns
numeric_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

# Preprocessing for categorical columns
categorical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

# Combine preprocessing
preprocessor = ColumnTransformer(
    transformers=[
```

```
        ("num", numeric_transformer, numerical_features),
        ("cat", categorical_transformer, categorical_features)
    ]
)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("\nTraining set shape:", X_train.shape)
print("Testing set shape:", X_test.shape)
```

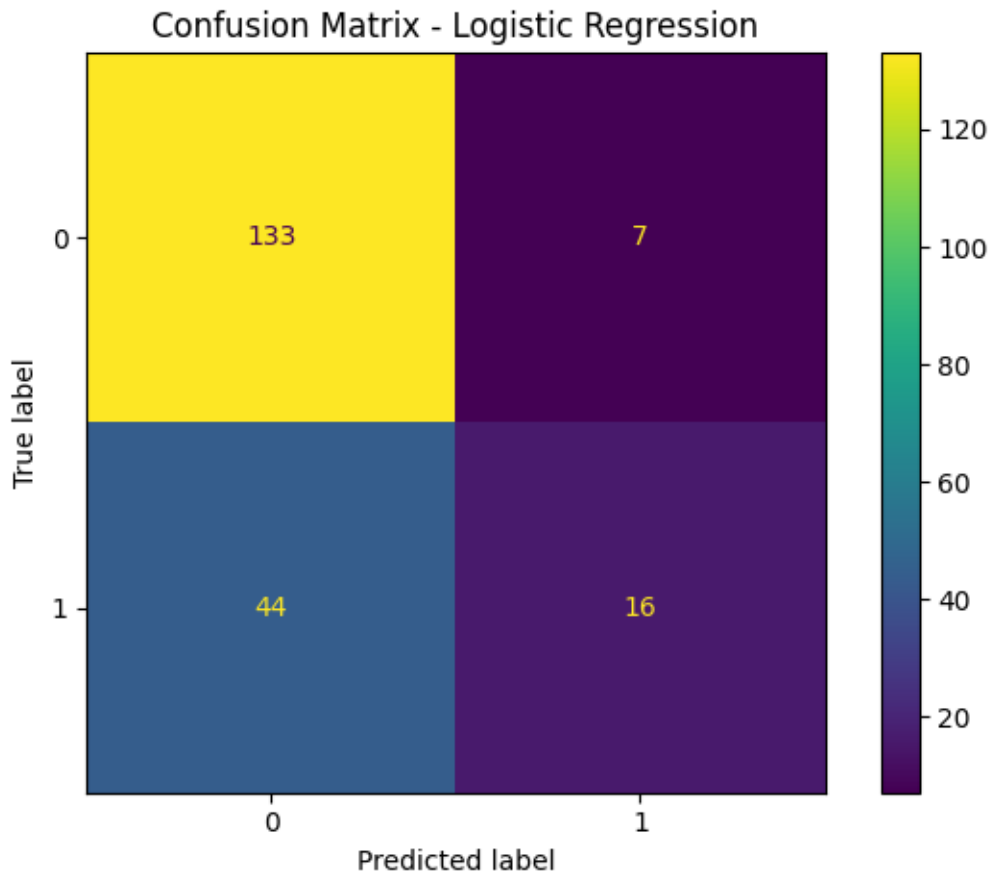
```
Mapped target values:
Risk
0    700
1    300
Name: count, dtype: int64

Categorical features:
['Sex', 'Housing', 'Saving accounts', 'Checking account', 'Purpose']

Numerical features:
['Age', 'Job', 'Credit amount', 'Duration']

Training set shape: (800, 9)
Testing set shape: (200, 9)
```

2.3 Logistic Regression Model

[11]:
```
log_model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("classifier", LogisticRegression(max_iter=1000, random_state=42))
])

log_model.fit(X_train, y_train)
y_pred_log = log_model.predict(X_test)

print("\n==============================")
print("Logistic Regression Results")
print("==============================")
print("Accuracy:", accuracy_score(y_test, y_pred_log))
print("\nClassification Report:")
```

```python
print(classification_report(y_test, y_pred_log))

cm_log = confusion_matrix(y_test, y_pred_log)
disp_log = ConfusionMatrixDisplay(confusion_matrix=cm_log)
disp_log.plot()
plt.title("Confusion Matrix - Logistic Regression")
plt.tight_layout()
plt.show()
```

```
==============================
Logistic Regression Results
==============================
Accuracy: 0.745

Classification Report:
              precision    recall  f1-score   support

           0       0.75      0.95      0.84       140
           1       0.70      0.27      0.39        60

    accuracy                           0.74       200
   macro avg       0.72      0.61      0.61       200
weighted avg       0.73      0.74      0.70       200
```

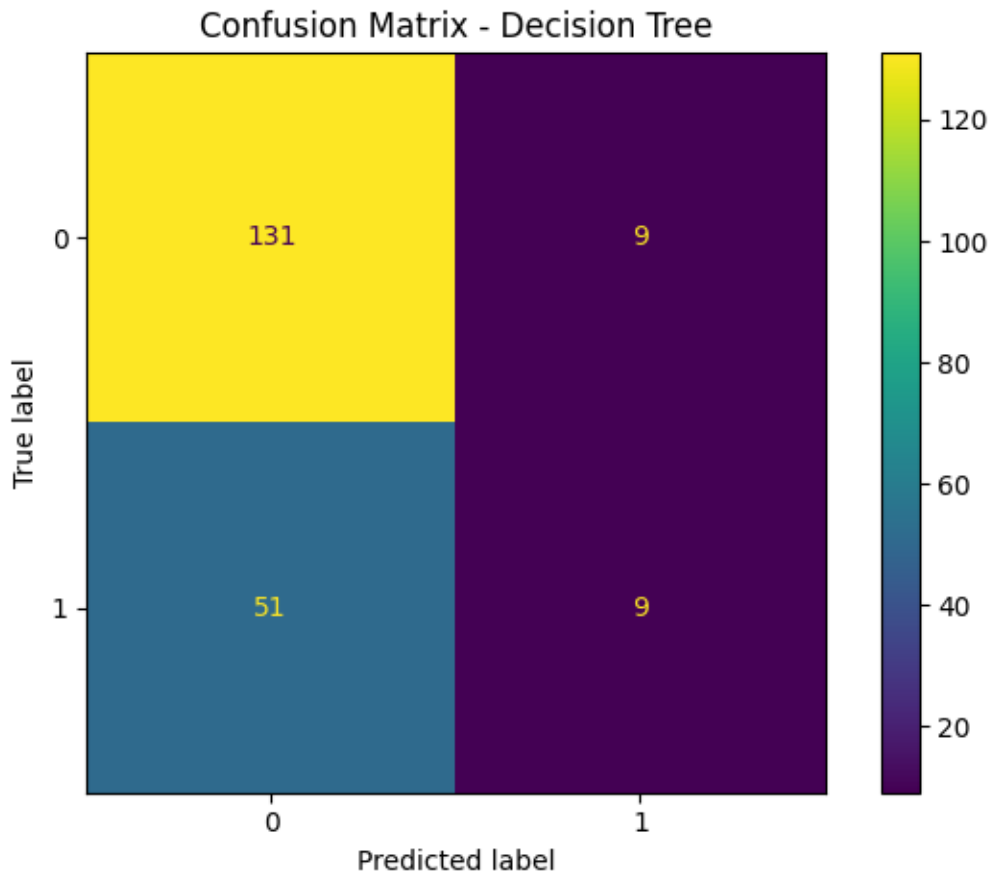


2.4 Decision Tree Model

[12]:
```
tree_model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("classifier", DecisionTreeClassifier(
        max_depth=5,
        random_state=42
    ))
])

tree_model.fit(X_train, y_train)
y_pred_tree = tree_model.predict(X_test)

print("\n==============================")
print("Decision Tree Results")
print("==============================")
print("Accuracy:", accuracy_score(y_test, y_pred_tree))
print("\nClassification Report:")
print(classification_report(y_test, y_pred_tree))
```

```python
cm_tree = confusion_matrix(y_test, y_pred_tree)
disp_tree = ConfusionMatrixDisplay(confusion_matrix=cm_tree)
disp_tree.plot()
plt.title("Confusion Matrix - Decision Tree")
plt.tight_layout()
plt.show()
```

```
==============================
Decision Tree Results
==============================
Accuracy: 0.7

Classification Report:
              precision    recall  f1-score   support

           0       0.72      0.94      0.81       140
           1       0.50      0.15      0.23        60

    accuracy                           0.70       200
   macro avg       0.61      0.54      0.52       200
weighted avg       0.65      0.70      0.64       200
```

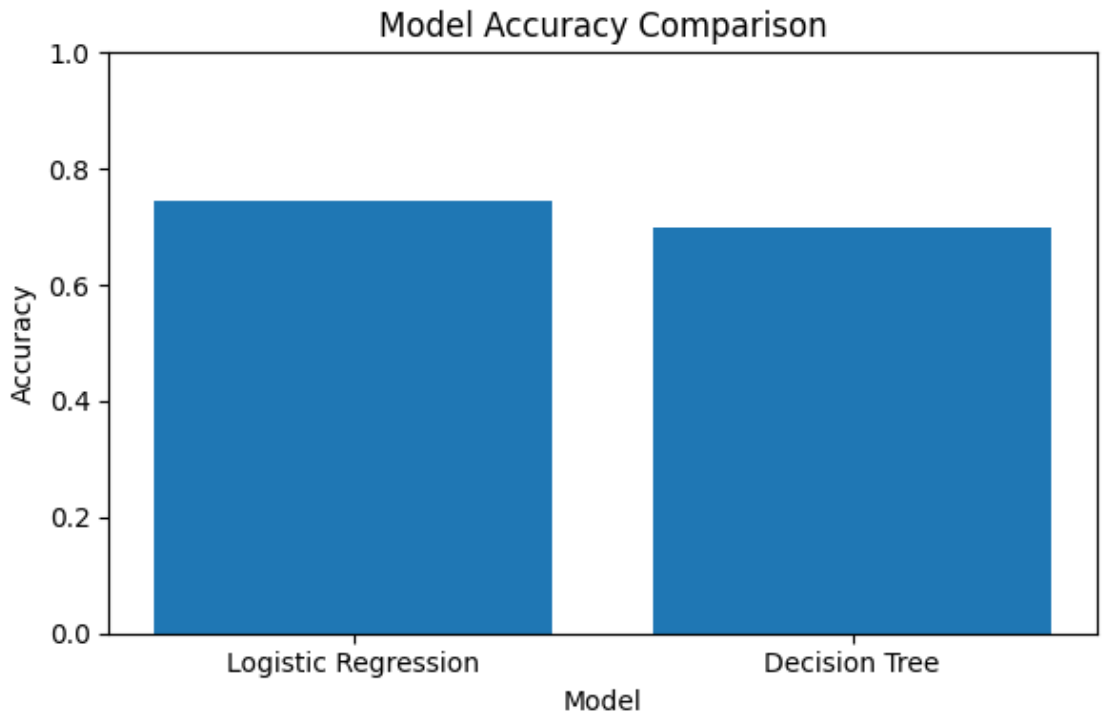


2.5 Compare model performance

[13]:
```
log_accuracy = accuracy_score(y_test, y_pred_log)
tree_accuracy = accuracy_score(y_test, y_pred_tree)

comparison_df = pd.DataFrame({
    "Model": ["Logistic Regression", "Decision Tree"],
    "Accuracy": [log_accuracy, tree_accuracy]
})

print("\nModel Comparison:")
display(comparison_df)

plt.figure(figsize=(6, 4))
plt.bar(comparison_df["Model"], comparison_df["Accuracy"])
plt.title("Model Accuracy Comparison")
plt.xlabel("Model")
plt.ylabel("Accuracy")
plt.ylim(0, 1)
```

```
plt.tight_layout()
plt.show()
```

```
Model Comparison:

                 Model  Accuracy
0  Logistic Regression     0.745
1        Decision Tree     0.700
```

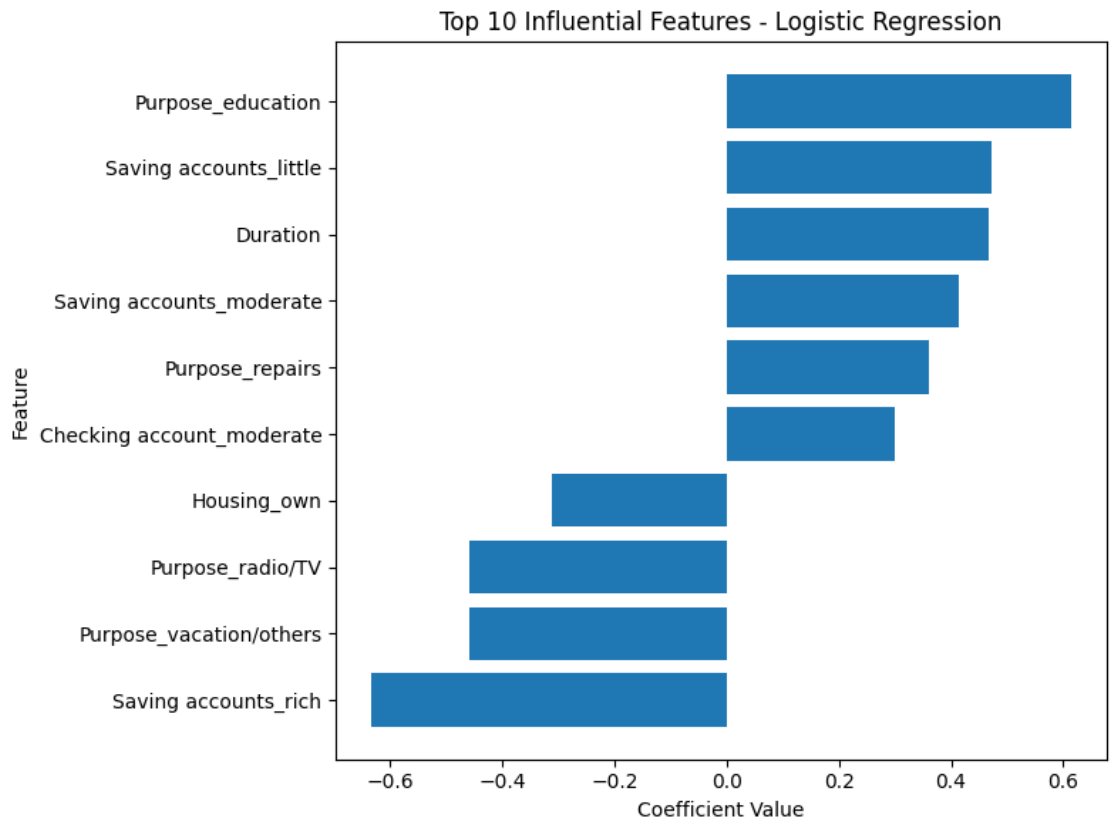


2.6 Feature importance from logistic regression

[14]:
```
# Get transformed feature names
ohe = log_model.named_steps["preprocessor"].named_transformers_["cat"].
 ↪named_steps["onehot"]
cat_feature_names = ohe.get_feature_names_out(categorical_features)

all_feature_names = np.concatenate([numerical_features, cat_feature_names])

# Logistic regression coefficients
coefficients = log_model.named_steps["classifier"].coef_[0]

coef_df = pd.DataFrame({
    "Feature": all_feature_names,
```

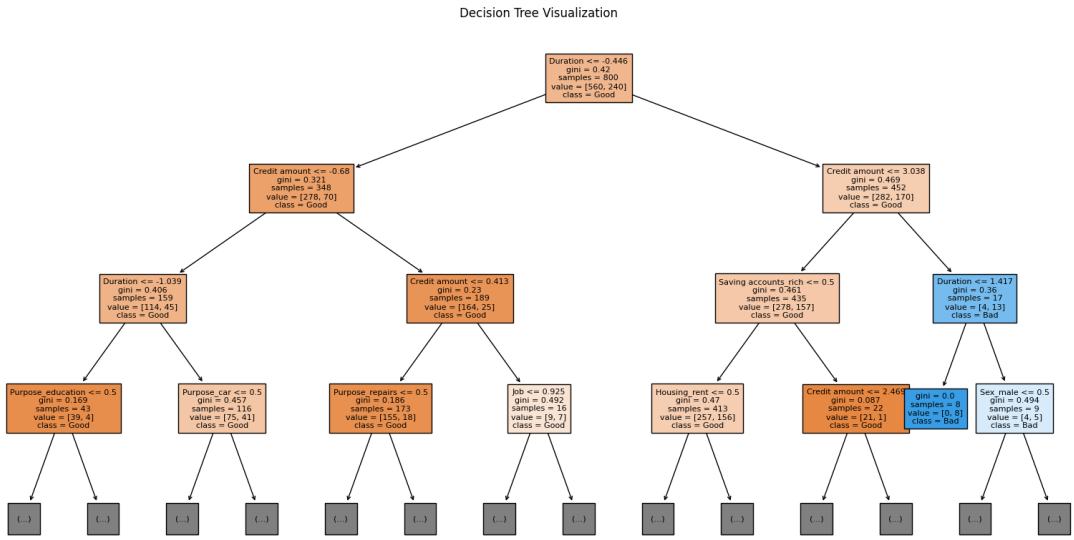
```
    "Coefficient": coefficients
})

coef_df["AbsCoefficient"] = coef_df["Coefficient"].abs()
coef_df = coef_df.sort_values("AbsCoefficient", ascending=False)

print("\nTop 10 most influential features (Logistic Regression):")
display(coef_df[["Feature", "Coefficient"]].head(10))

top_10 = coef_df.head(10).sort_values("Coefficient")

plt.figure(figsize=(8, 6))
plt.barh(top_10["Feature"], top_10["Coefficient"])
plt.title("Top 10 Influential Features - Logistic Regression")
plt.xlabel("Coefficient Value")
plt.ylabel("Feature")
plt.tight_layout()
plt.show()
```

```
Top 10 most influential features (Logistic Regression):

                      Feature  Coefficient
12       Saving accounts_rich    -0.634708
19          Purpose_education     0.615170
9      Saving accounts_little     0.471012
3                    Duration     0.466022
23    Purpose_vacation/others    -0.459763
21           Purpose_radio/TV    -0.458286
10   Saving accounts_moderate     0.414423
22            Purpose_repairs     0.360532
7                 Housing_own    -0.311498
14  Checking account_moderate     0.298841
```



2.7 visualize a small decision tree

[15]:
```
# To plot the tree, we need the transformed data
X_train_processed = tree_model.named_steps["preprocessor"].transform(X_train)

# For plotting feature names again
tree_feature_names = all_feature_names

plt.figure(figsize=(20, 10))
plot_tree(
    tree_model.named_steps["classifier"],
    filled=True,
    feature_names=tree_feature_names,
    class_names=["Good", "Bad"],
    max_depth=3,
    fontsize=8
)
plt.title("Decision Tree Visualization")
plt.show()
```



2.8 Final interpretation summary

[16]:
```
best_model = "Logistic Regression" if log_accuracy >= tree_accuracy else␣
↪"Decision Tree"

print("\n========================================")
print("Final Summary")
print("========================================")
print(f"Best performing model based on accuracy: {best_model}")
print(f"Logistic Regression Accuracy: {log_accuracy:.4f}")
print(f"Decision Tree Accuracy: {tree_accuracy:.4f}")

print("\nProject conclusion:")
print(
    "This analysis shows that machine learning can be used to classify credit␣
↪risk "
    "using applicant financial and demographic information. The results provide␣
↪a useful "
    "foundation for understanding borrower risk and supporting loan approval␣
↪decisions."
)
```

```
========================================
Final Summary
========================================
Best performing model based on accuracy: Logistic Regression
Logistic Regression Accuracy: 0.7450
```

```
Decision Tree Accuracy: 0.7000

Project conclusion:
This analysis shows that machine learning can be used to classify credit risk
using applicant financial and demographic information. The results provide a
useful foundation for understanding borrower risk and supporting loan approval
decisions.
```